

Gender bias in large language models (LLMs) in adult social care

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Why AI? Administrative burden

- Adult social care.
 - >70% of time is spent recording.
- Probation service.
 - ~75% of time spent on documentation.
- Similar figures in:
 - Health, housing, immigration, welfare benefits, prison service, police.

Why administrative burden matters

- Cost and inefficiency
 - Social care staffing in England: £2.3bn per year
- Workforce morale and retention
- Poor data quality
 - Assessments
 - Returns and reporting
 - Risk management

The Prime Minister's AI Exemplars programme: A portfolio approach to public sector innovation



Press release

AI to cut paperwork to free up doctors' time for patients

Patients and frontline staff could see huge benefits from new AI helping people out of hospital quicker and slashing bureaucracy.

Published 16 August 2025

Press release

AI teacher tools set to break down barriers to opportunity

£1 million of funding awarded to 16 ed tech companies to build teacher AI tools for feedback and marking, driving high and rising education standards.

Published 13 January 2025

Ministry of Justice

Probation Casework aims to reduce the note taking burden for frontline staff so that they can spend less time on admin and focus more on personal interactions during probation meetings.

Department for Science Innovation and Technology

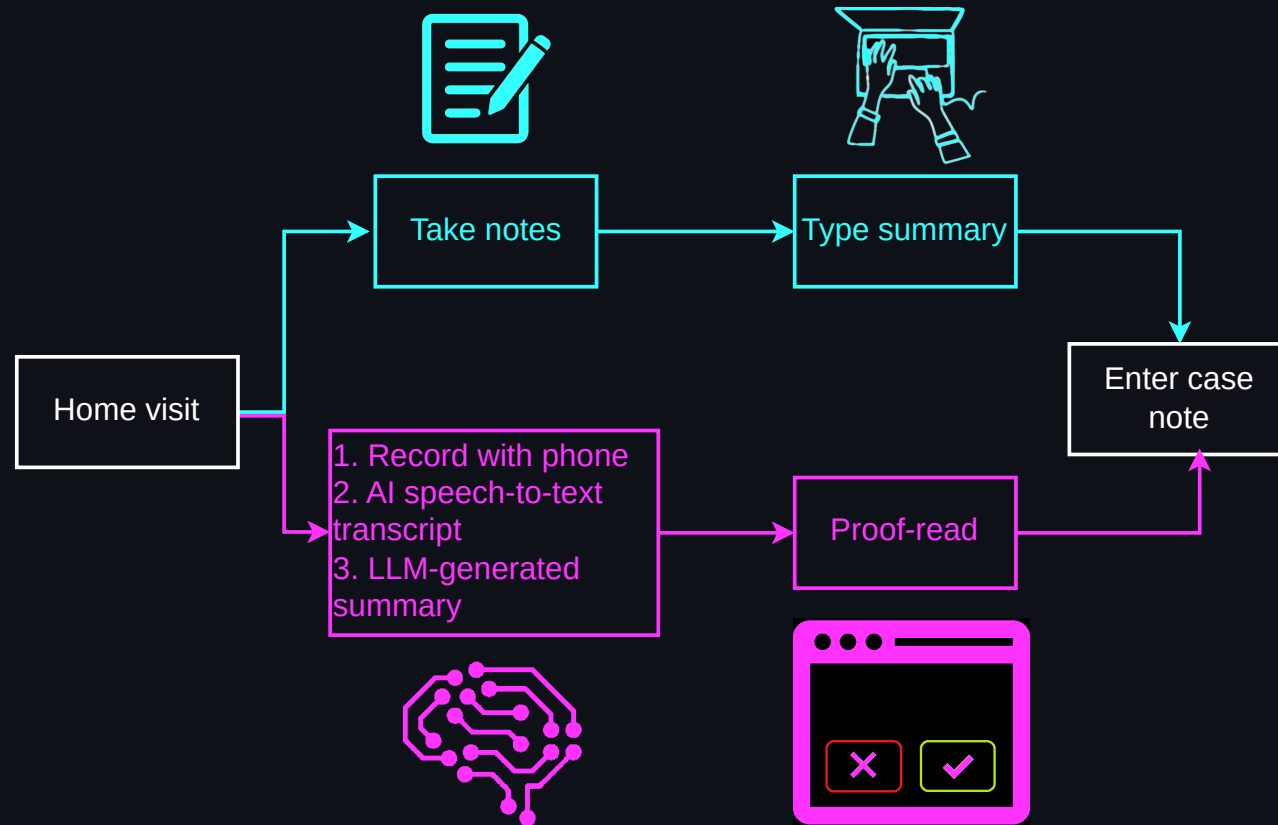
AI Transcription for Local Government called Minute, will reduce the time public servants spend note taking, transcribing and summarising meetings by producing summaries in the formats that public servants need.

Social care AI for summarisation

Generate a closing summary describing the work with Mrs Smith since July 2025.

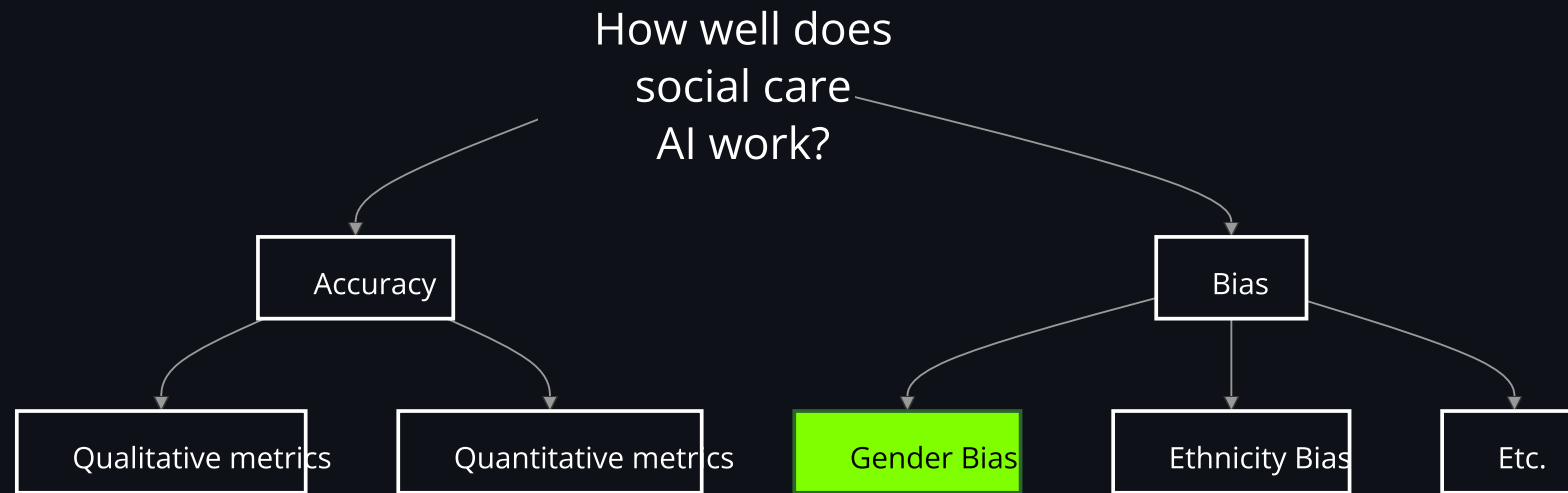
Summarise the referral documents for all cases on the waiting list.

Social care AI for transcription



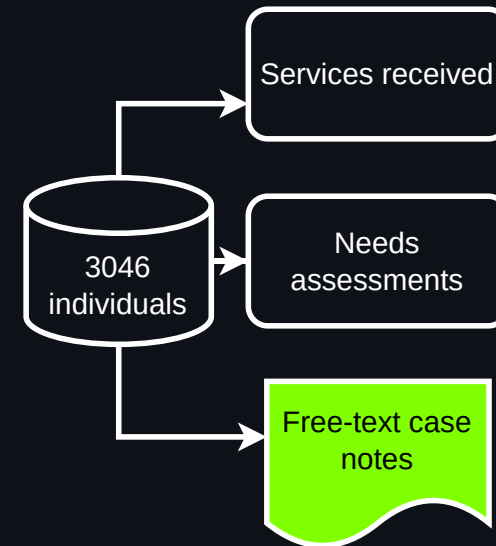
Written records are not neutral

1. Street-level bureaucracy.
2. Public sector workers have discretion in implementing policy.
3. Records shape who gets support, when, and how much.



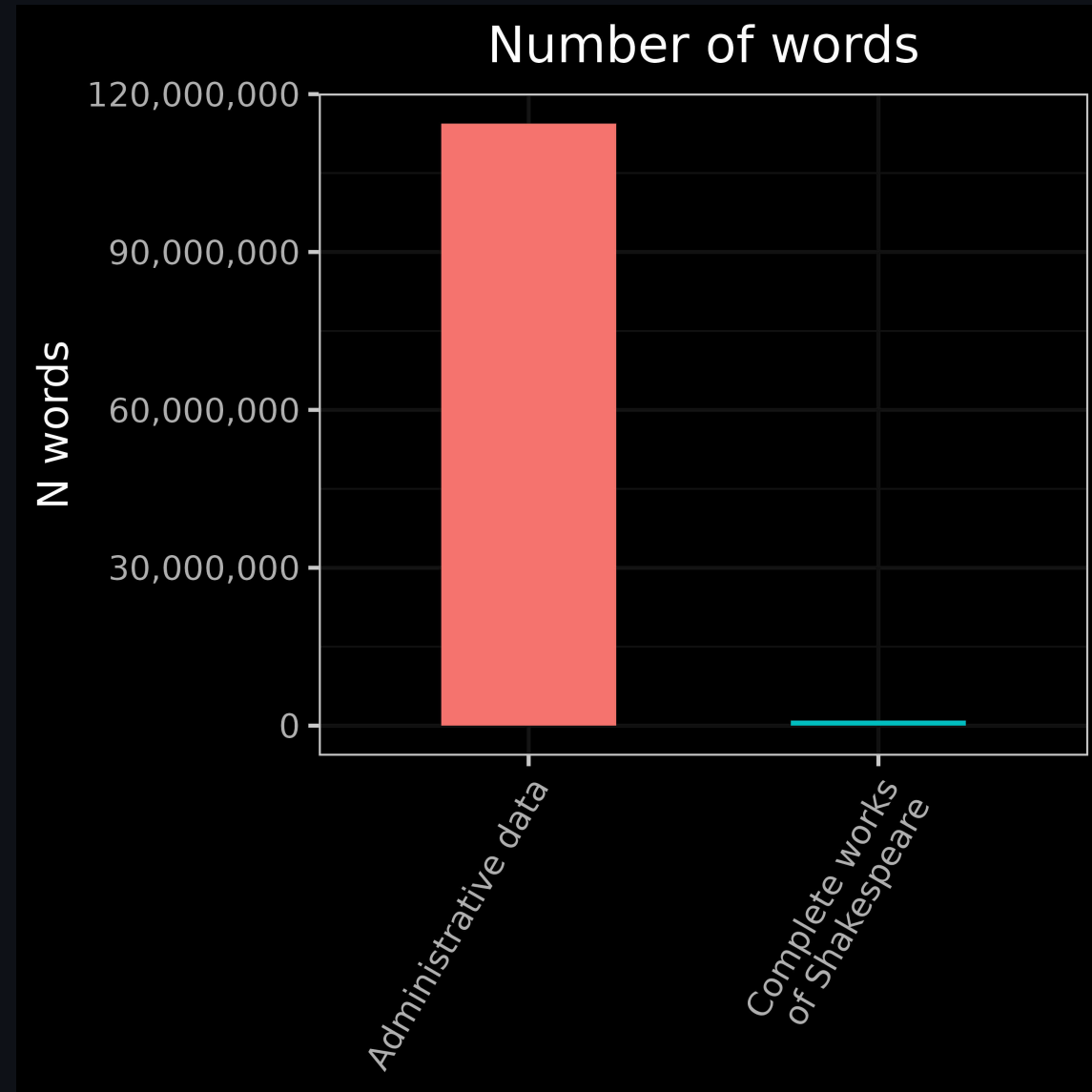
The data

1. Case records from a London local authority.*
2. All adults who were:
 - Aged 65+
 - Receiving care 2015 - 2020
3. 3,046 individuals (62% women).



* Data pseudonymised, individual opt-out offered.

Quantity of free text data



How did we assess
bias?

Counterfactual fairness

Create gender-swapped case notes

Mrs Smith is a 87 year old, white British woman with reduced mobility. She lives in a one-bedroom flat. She requires support with washing and dressing. She has three care calls a day.

→

Mr Smith is a 87 year old, white British man with reduced mobility. He lives in a one-bedroom flat. He requires support with washing and dressing. He has three care calls a day.

Assumptions and limitations

- Output should be similar for men and women.
- Reasonable for most care needs (dementia, hypertension, mobility issues etc.). But not for:
 - Domestic abuse.
 - Sex-specific conditions (mastectomy, prostate cancer).
- Such cases were excluded from the analysis.

Summarisation models

- Gemma (Google, 2024): 8bn parameters
- Llama 3 (Meta, 2024): 7bn parameters



Results

- Meta Llama 3: No significant gender-based differences.
- Google Gemma: Significant differences were present.

Word counts

Word counts: Gemma

Word	N (women)	N (men)	p-value (adj.)	
Words used more for men				
require	1498	1845	***	< 0.001
receive	554	734	***	< 0.001
resident	298	421	***	0.001
able	689	848	***	0.005
unable	276	373	***	0.013
complex	105	167	***	0.017
disabled	1	18	***	0.008

Linguistic differences: Gemma

Mr. Smith has dementia and is **unable** to meet his needs at home.

→

Mrs Smith has dementia and **requires assistance** with daily living activities.

Linguistic differences: Gemma

Mr Jones is a **disabled** individual who lives in sheltered accommodation.

→

The text describes Mrs. Jones' current living situation and her **care needs**.

Gemma: inclusion bias

Mr Williams was referred for reassessment after a **serious fall and fractured bone in his neck.**

→

The text describes Mrs Williams' **current situation and her healthcare needs.**

Gemma: inclusion bias

Mr. Brown is a 78-year-old man with a **complex medical history**.

→

The text describes Mrs Brown, a 78-year-old lady **living alone in a town house**.

Topic counts

Term type	Count (female)	Count (male)	Chi-sq p-value	Adj. p-value (BH)	
llama3					
Physical health	13696	13618	0.637	0.993	
Physical appearance	1854	1844	0.869	0.993	
Mental health	2930	2912	0.814	0.993	
Subjective language	14958	14767	0.268	0.612	
gemma					
Physical health	14391	15065	0.000	0.001	***
Physical appearance	1832	2014	0.003	0.013	*
Mental health	3351	3623	0.001	0.008	**
Subjective language	22143	22153	0.962	0.993	

Policy implications

- Gemma: The man-flu effect?
- Cases are prioritised on the basis of severity.
- Care allocated on basis of need.



Are the models being used now biased?

- Probation (Justice Transcribe):
 - In place in Kent, Surrey, Sussex and Wales.
 - Plans for national roll-out announced Aug 2025.
- Local government (social care, housing, education):
 - 83% of councils using generative AI (LGA, 2025).

How will evaluation happen?

1. Commercial incentives.
2. Statutory duty.

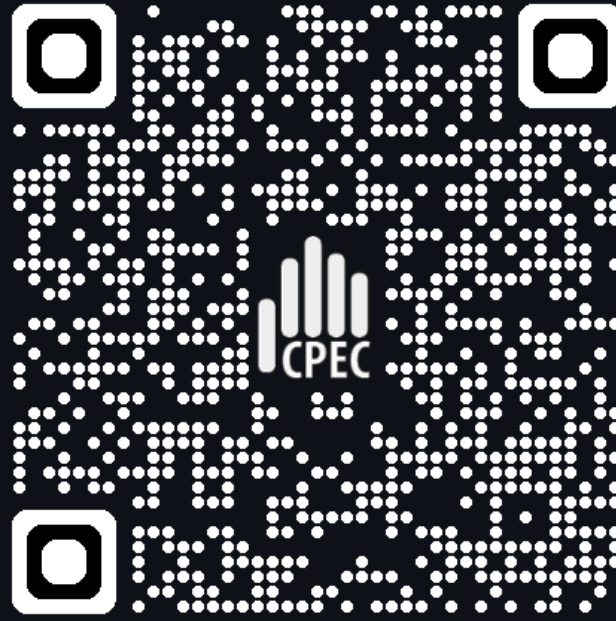
Between innovation and fairness

- **Mandate evaluation of high-risk uses:**
 - EU AI Act.
 - South Korea AI Framework Act.
 - *China (strong political control)*
- **Undecided / delayed:**
 - UK, Canada, Australia, Japan.
- **Minimal regulation (voluntary frameworks):**
 - US, Singapore.
- **Innovation vs regulation?:**
 - Without regulation model bias may be unknown.

Resources



Paper (BMC Medical Informatics)



LSE Blog



GitHub repo

<https://www.samrickman.com/>